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Course Code: ITA0305

Course Name: Mobile Computing

Experiment 3: Call Blocking Probability using the Erlang B Formula

Program Code:

```
import numpy as np
import matplotlib.pyplot as plt
# Erlang B formula
def erlang_b(A, C):
    B = 1.0
    for i in range(1, C + 1):
        B = (A * B) / (i + A * B)
    return B
A = np.arange(1, 21)      # Traffic intensity in Erlangs
C = 10                    # Number of channels
blocking = [erlang_b(a, C) for a in A]
# Display output
print("----- Call Blocking Probability -----")
print("Traffic(Erlang)  Blocking Probability(%)")

for a, b in zip(A, blocking):
    print(f"{a:<16} {b*100:.2f}")
print("\nNumber of Channels:", C)
print("Result: Call blocking probability calculated successfully.")
# Plot graph
```

```
plt.plot(A, np.array(blocking)*100, "o-", label="Erlang B")

plt.xlabel("Traffic Intensity (Erlangs)")

plt.ylabel("Blocking Probability (%)")

plt.title("Call Blocking Probability using Erlang B Formula")

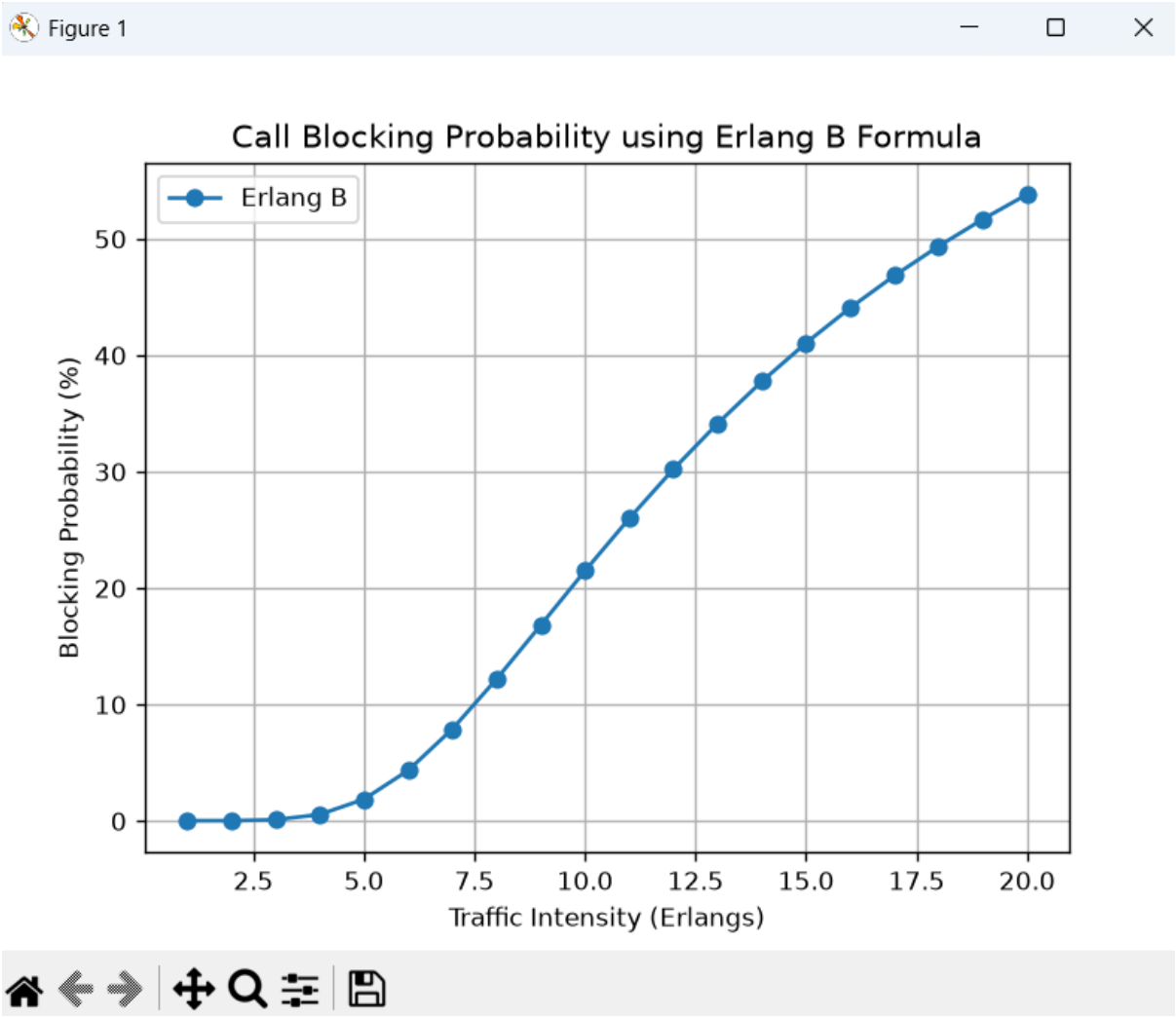
plt.legend()

plt.grid(True)

plt.show()
```

```
======== RESTART: D:/knowledge_Engineering/mobile computing/
----- Call Blocking Probability -----
Traffic(Erlang)  Blocking Probability(%)
1                0.00
2                0.00
3                0.08
4                0.53
5                1.84
6                4.31
7                7.87
8                12.17
9                16.80
10               21.46
11               25.96
12               30.19
13               34.12
14               37.73
15               41.03
16               44.06
17               46.82
18               49.35
19               51.67
20               53.80

Number of Channels: 10
Result: Call blocking probability calculated successfully.
```


Figure 1
Call Blocking Probability using Erlang B Formula
Erlang B
Blocking Probability (%)
0
10
20
30
40
50
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Traffic Intensity (Erlangs)